Software Requirements Specification

Open Clinical Record

Open Clinical Record Internship Project

Requirements Baseline Draft — September 2026

Document Status: Draft for Mentor and Clinical Review

Document Item	Value
Product	Open Clinical Record
Document	Software Requirements Specification (SRS)
Version	1.1 — detailed academic baseline
Status	Draft for review and requirements baseline
Primary scope	Patient chart, medical records/reports, appointments and encounter workflow
Primary users	Clinician/Doctor; Nurse/Clinical Staff; Receptionist/Front Desk
Evidence status	Preliminary clinical validation from one clinician survey response; broader validation remains required
Next lifecycle stage	Architecture and data-model design after requirements review

This document is a requirements specification. It defines required system behavior and quality characteristics without committing the implementation to a particular programming language, framework, database engine, or deployment architecture unless explicitly identified as a constraint.

Contents

1 Introduction

1.1 Purpose of the Requirements Document

This Software Requirements Specification defines the functional behavior, quality requirements, interfaces, constraints, data requirements, business rules, and acceptance expectations for the Open Clinical Record (OCR) MVP. Its purpose is to establish a clear and testable baseline before architecture, detailed data modeling, and implementation are finalized.

The document translates the project's research and clinical workflow discovery into explicit requirements. It is intentionally more detailed than a feature list: each requirement is written so that a future developer, tester, mentor, or clinical reviewer can determine what the system is expected to do and whether the expectation has been satisfied.

The SRS also establishes traceability. Requirements are connected to the clinical workflow and business rules and will subsequently be connected to architecture decisions, data structures, API contracts, implementation tasks, and verification evidence. A requirement appearing in this document does not by itself mean that the requirement has already been implemented or clinically validated.

1.2 Scope of Product

Open Clinical Record is an extensible Electronic Medical Record platform for managing a focused set of common clinical workflows. The MVP centers on three connected capabilities:

1. maintaining a longitudinal patient chart;
2. managing appointments, visits, check-in, and basic walk-in handling; and
3. recording clinical encounters, medical records, and clinical reports with appropriate document integrity.

The product is deliberately not defined as a complete hospital information system. Laboratory, pharmacy, billing, insurance, radiology, patient portal, advanced analytics, AI-assisted clinical decision support, and dedicated mobile applications are future expansion areas unless later approved through requirements change control.

1.3 Definitions and Abbreviations

Term	Definition
EMR	Electronic Medical Record; a digital record of patient information and clinical care maintained within a healthcare organization or clinical system.
SRS	Software Requirements Specification; the controlled document defining system requirements.
MVP	Minimum Viable Product; the smallest planned release that satisfies the defined initial clinical scope.
Patient chart	The longitudinal clinical context for a patient, including identity, demographics, status, history, encounters, appointments, and relevant clinical information.

Appointment	A scheduled interaction between a patient and a provider or service at a defined date/time.
Visit	The operational occurrence in which a patient attends a healthcare service; a visit may originate from an appointment or a walk-in.
Encounter	The clinical interaction during which assessment, examination, diagnosis, treatment, prescription, and related documentation may be recorded.
Clinical document	A clinical note, medical report, or other formal record documenting care.
Finalized document	A clinical document whose content has been completed and protected from ordinary editing.
Amendment	An authorized correction or addition to a finalized document that preserves the original content and records the change.
Walk-in	A patient visit initiated without a previously scheduled appointment.
RBAC	Role-Based Access Control; access decisions based on an authenticated user's assigned role and permissions.
Audit trail	A chronological record of significant system and clinical-record actions, including actor and time information.
FHIR	Fast Healthcare Interoperability Resources, an HL7 standard relevant to future interoperability.
API	Application Programming Interface used to expose or consume software functionality.
UI	User Interface.
UX	User Experience.
NFR	Non-Functional Requirement.
CRUD	Create, Read, Update, Delete operations.

1.4 References

The requirements baseline is informed by the project's internal research and discovery artifacts, including the clinical workflow, patient-chart research, appointment-scheduling research, clinical-documentation research, research log, and architecture decision record. The project also considers established healthcare interoperability concepts such as HL7 FHIR as a future-facing boundary rather than an MVP integration requirement.

Primary internal references:

- `docs/02-discovery/clinical-workflow.md`
- `docs/01-research/patient-chart-research.md`
- `docs/01-research/appointment-scheduling-research.md`
- `docs/01-research/clinical-documentation-research.md`
- `docs/01-research/research-log.md`
- `docs/04-architecture/adr/0001-record-architectural-decisions.md`
- `docs/03-requirements/traceability-matrix.md`

1.5 Overview of the Remainder of the Document

Section 2 describes the product from a general perspective, including its environment, major functions, users, constraints, assumptions, and dependencies. Section 3 defines the specific requirements, including external interfaces, functional requirements, non-functional requirements, logical database requirements, design constraints, and quality attributes. Section 4 contains supporting appendices such as data structures, workflow models, domain concepts, traceability, validation evidence, and open questions. Section 5 provides an index of important requirements concepts and identifiers.

2 General Description

2.1 Product Perspective

Open Clinical Record is conceived as an extensible EMR platform rather than a collection of unrelated CRUD screens. The patient chart is the principal clinical context. Appointments and encounters are linked to the patient, and clinical documentation is linked to the encounter and patient so that the longitudinal record remains coherent.

The intended high-level workflow is:

Patient Registration/Search → Chart → Appointment/Visit → Check-in → Encounter → Clinical Documentation → Longitudinal Record

The MVP may be deployed in a relatively small clinical environment. The requirements therefore emphasize correctness, privacy, maintainability, clear role boundaries, and resilience to practical connectivity problems without claiming the operational guarantees of a large national health information exchange.

2.2 Product Functions

The major product functions are:

1. Patient registration and unique identification.
2. Patient search and chart access.
3. Clinical summary, allergies, medications, alerts, and patient status.
4. Appointment creation, viewing, rescheduling, cancellation, history, and check-in.
5. Flexible walk-in handling.
6. Patient-to-visit-to-encounter linkage.
7. Clinical encounter documentation: complaint, vitals, history, examination, diagnosis, treatment, prescription, and notes.
8. Medical report generation and document lifecycle management.
9. Finalization and controlled amendment of clinical documents.
10. Authentication, role-based authorization, auditability, and privacy protection.
11. Input validation and safe error handling.

2.3 User Characteristics

The MVP has exactly three primary application roles.

Role	Typical responsibility	Primary system capabilities
Receptionist / Front Desk	Patient-facing administrative workflow	Register and search patients; manage permitted demographic information; create, view, reschedule, and cancel appointments; perform check-in; initiate supported walk-in intake. No diagnosis, treatment, prescribing, or clinical-report authoring.
Nurse / Clinical Staff	Clinical workflow support and nursing activities	Patient search; appointment/check-in support; record permitted preliminary clinical information such as vitals; support encounter workflow; access clinical information required for assigned duties. Clinical privileges remain permission-controlled.
Clinician / Doctor	Clinical assessment and clinical decision documentation	Review chart; assess patient; record history and examination; diagnose; document treatment; prescribe when applicable; create/finalize clinical notes and reports; perform authorized patient-status changes and amendments.

A role is not equivalent to unrestricted access. Permissions shall be enforced at the operation and data level. The exact final permission matrix must be reviewed with the mentor/clinical stakeholder before implementation is considered complete.

2.4 General Constraints

- The MVP shall remain within the defined patient chart, appointment/visit, and medical record/report scope.
- Patient and clinical information is sensitive and shall be protected against unauthorized access.
- The system shall preserve clinically significant history rather than silently destroying it.
- Finalized clinical documents shall have stronger integrity protection than ordinary draft content.
- The solution should use clear application boundaries so future interoperability can be added without redesigning the clinical domain from scratch.
- The system shall not make clinical decisions autonomously in the MVP.
- Walk-in workflow shall remain flexible because urgency and patient condition may change the appropriate operational path.

2.5 Assumptions and Dependencies

ID	Assumption / dependency
A-001	Users are authenticated before protected clinical information is displayed.
A-002	The deployment has access to a supported persistent data store.
A-003	Patient identifiers are unique within the system's operational scope.
A-004	Clinical roles and permissions are approved by the responsible project/clinical stakeholder.
A-005	The organization using the system defines applicable retention, privacy, and access policies.
A-006	Date and time values are recorded consistently and are interpretable in the deployment's configured timezone.
A-007	Future FHIR interoperability, if required, will be based on the finalized domain/data model rather than being assumed to be part of the MVP.
A-008	Network connectivity may occasionally fail; critical write operations therefore require safe failure behavior.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface Requirements

The application shall provide a consistent interface for the three approved roles. The UI shall make the patient's identity and current context clear when working with clinical information. Screens shall distinguish editable drafts from finalized documents and shall present appointment status explicitly.

The interface shall support, at minimum, the following conceptual views:

- authentication/sign-in;
- patient search and registration;
- patient chart and clinical summary;
- appointment list/calendar and appointment detail;
- check-in and walk-in intake;
- encounter workspace;
- clinical documentation/report workspace;
- finalized-document and amendment history view; and
- appropriate error, validation, and access-denied states.

The UI shall not expose controls for actions the current user is not permitted to perform. Where hiding a control is insufficient for security, the server/application service layer shall independently enforce the permission.

3.1.2 Software Interfaces

The system shall be designed around explicit software boundaries so that the UI does not directly manipulate clinical persistence structures. At minimum, the conceptual boundary shall distinguish authentication/authorization, patient management, scheduling, encounter/documentation, persistence, and audit concerns.

If an API layer is used, endpoints shall represent domain operations rather than exposing unrestricted database CRUD. Future FHIR mappings should be possible at the integration boundary without making internal persistence identical to FHIR resources.

3.1.3 Communication Interfaces

Communication between client and server components shall use a secure transport mechanism appropriate to the deployment. Authentication credentials and protected clinical information shall not be transmitted through unprotected channels in a production deployment.

The system shall provide understandable feedback when a communication failure prevents an operation from completing. It shall not report an operation as successfully saved when persistence has not been confirmed.

3.2 Functional Requirements

3.2.1 Patient Management Requirements

ID	Requirement	Detailed specification
FR-PAT-001	**Register patient**	The system shall allow an authorized Receptionist/Front Desk or other permitted role to create a patient record containing the required demographic and administrative information.
FR-PAT-002	**Assign patient identifier**	The system shall assign or record a unique patient identifier and shall reject duplicate identifiers where uniqueness is required.
FR-PAT-003	**Search patient**	The system shall allow authorized users to search using supported identifying information and shall present results in a way that reduces accidental selection of the wrong patient.
FR-PAT-004	**View patient profile**	The system shall display demographic information and current patient status to users with appropriate permission.
FR-PAT-005	**View clinical summary**	The system shall provide a concise longitudinal summary including available conditions, previous diagnoses, allergies, medications, relevant vitals, and previous visits/encounters.
FR-PAT-006	**Record allergies**	The system shall allow appropriately authorized clinical users to record and update allergy information.
FR-PAT-007	**Record medications**	The system shall allow appropriately authorized users to record relevant medication information and associate it with the correct patient.

FR-PAT-008	**Display alerts**	The system shall surface clinically relevant alerts, including recorded allergies, critical conditions, medication/adverse-reaction risks, follow-up items, and precautions where applicable.
FR-PAT-009	**Record patient status**	The system shall maintain a current patient status and shall restrict status changes according to role permissions.
FR-PAT-010	**Mark deceased patient**	The system shall allow an authorized Clinician/Doctor to record deceased status together with the date of death and required supporting information.
FR-PAT-011	**Handle deceased patient**	After deceased status is recorded, the system shall prevent inappropriate future appointment activity and shall flag or cancel affected future appointments according to the approved policy.

3.2.2 Appointment and Visit Requirements

ID	Requirement	Detailed specification
FR-APT-001	**Create appointment**	The system shall allow an authorized Receptionist/Front Desk or Nurse/Clinical Staff user, where permitted, to create an appointment with patient, date, time, service/provider, and required scheduling information.
FR-APT-002	**View appointments**	The system shall allow authorized users to view appointments relevant to their role and permissions.
FR-APT-003	**Reschedule appointment**	The system shall allow an authorized user to reschedule an eligible appointment while preserving the original scheduling history.
FR-APT-004	**Cancel appointment**	The system shall allow an authorized user to cancel an eligible appointment without deleting its historical existence.
FR-APT-005	**Preserve appointment history**	Cancelled and rescheduled appointments shall remain distinguishable from active appointments and shall retain sufficient status/history information for later review.
FR-APT-006	**Check in patient**	The system shall allow authorized Receptionist/Front Desk or Nurse/Clinical Staff users to record check-in for an eligible appointment or supported walk-in.
FR-APT-007	**Support walk-in workflow**	The system shall support a walk-in path that can adapt to urgency and patient condition rather than requiring every walk-in to become an ordinary scheduled appointment.
FR-APT-008	**Link appointment to encounter**	When an encounter follows an appointment, the system shall preserve the association between the appointment, visit, encounter, and patient.

FR-APT-009	**Prevent deceased-patient booking**	The system shall prevent inappropriate future appointments for patients whose status is deceased.

3.2.3 Encounter and Clinical Assessment Requirements

ID	Requirement	Detailed specification
FR-ENC-001	**Start encounter**	The system shall allow an authorized clinical user to start or access an encounter associated with a patient visit.
FR-ENC-002	**Record chief complaint**	The system shall allow the Clinician/Doctor to record the patient's presenting complaint or reason for the visit.
FR-ENC-003	**Record vitals**	The system shall allow Nurse/Clinical Staff, and other explicitly authorized clinical users, to record relevant vital signs.
FR-ENC-004	**Record history**	The system shall allow the Clinician/Doctor to document relevant medical history and encounter history.
FR-ENC-005	**Record examination**	The system shall allow the Clinician/Doctor to document examination findings.
FR-ENC-006	**Record diagnosis**	The system shall allow only appropriately authorized Clinician/Doctor users to record the diagnosis or clinical assessment.
FR-ENC-007	**Record treatment**	The system shall allow appropriately authorized Clinician/Doctor users to document treatment and care plans.
FR-ENC-008	**Record prescription**	The system shall allow appropriately authorized Clinician/Doctor users to record prescription information where applicable.
FR-ENC-009	**Create clinical note**	The system shall allow an authorized Clinician/Doctor to create and save clinical notes associated with the correct encounter.
FR-ENC-010	**Maintain encounter linkage**	Clinical encounter information shall remain linked to exactly the intended patient and encounter context and shall not be silently reassigned.

3.2.4 Medical Records and Reports Requirements

ID	Requirement	Detailed specification
FR-REC-001	**Generate medical report**	The system shall allow an authorized Clinician/Doctor to generate a medical report using relevant patient and encounter information.

FR-REC-002	**Finalize clinical document**	The system shall allow an authorized Clinician/Doctor to finalize an eligible clinical document.
FR-REC-003	**Restrict finalized editing**	A finalized document shall not be ordinarily editable in a way that silently changes its finalized content.
FR-REC-004	**Amend finalized document**	The system shall provide an authorized correction/amendment path for finalized documents according to the approved clinical policy.
FR-REC-005	**Preserve original document**	When an amendment occurs, the original finalized content shall remain preserved and identifiable.
FR-REC-006	**Record amendment history**	The system shall record amendment metadata including responsible user, date/time, reason or explanation where required, and relationship to the original document.
FR-REC-007	**View document status**	The system shall clearly distinguish draft, finalized, amended, and other approved lifecycle states.

3.2.5 Authentication, Authorization and Audit Requirements

ID	Requirement	Detailed specification
FR-SEC-001	**Authenticate user**	The system shall authenticate users before granting access to protected application functions or clinical data.
FR-SEC-002	**Enforce role permissions**	The system shall enforce permissions associated with the three approved roles: Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
FR-SEC-003	**Restrict clinical actions**	Diagnosis, treatment, prescriptions, clinical notes, medical reports, patient-status changes, and finalized-document amendments shall require appropriate authorization.
FR-SEC-004	**Audit sensitive actions**	The system shall record significant actions affecting patient data, appointments, clinical documents, permissions, and other security-sensitive operations.
FR-SEC-005	**Protect unauthorized access**	Requests that lack the required permission shall be denied without exposing protected data.

3.2.6 Validation and Error Handling Requirements

ID	Requirement	Detailed specification
FR-VAL-001	**Validate required data**	The system shall validate required fields, formats, relationships, and applicable business rules before committing records.

FR-VAL-002	**Prevent invalid appointments**	The system shall reject appointment operations that violate patient status, scheduling, or other approved rules.
FR-VAL-003	**Report failures clearly**	The system shall provide understandable error feedback while avoiding disclosure of unnecessary sensitive information.
FR-VAL-004	**Preserve data on failure**	A failed save or communication operation shall not silently create partial clinical records or falsely report success.

3.3 Business Rules

ID	Business rule
BR-001	Cancelled appointments remain in history and are clearly marked as cancelled.
BR-002	Rescheduled appointments retain sufficient history to distinguish the original appointment from the updated scheduling state.
BR-003	Finalized clinical documents cannot be silently overwritten; corrections are represented as traceable amendments while preserving the original.
BR-004	Diagnosis, treatment, prescription, clinical notes, reports, and patient-status changes are protected by role-based permissions.
BR-005	Clinically relevant alerts, including allergies and high-priority risks, are displayed in the appropriate clinical context.
BR-006	Deceased patients cannot receive inappropriate future appointments; existing future appointments are handled according to the configured policy.
BR-007	Walk-in handling remains flexible because urgency and patient condition may affect the appropriate workflow.
BR-008	Clinical information remains associated with the correct patient and encounter.
BR-009	A user shall not gain additional clinical authority merely by accessing a patient chart.
BR-010	Historical information shall not be silently destroyed when an operational state changes.

3.4 Non-Functional Requirements

3.4.1 Security and Privacy

ID	Requirement
NFR-SEC-001	The system shall require authenticated access to protected clinical data.
NFR-SEC-002	Authorization shall be enforced for protected operations and shall not depend solely on UI visibility.

NFR-SEC-003	Access shall follow least-privilege principles within the approved three-role model.
NFR-SEC-004	Sessions shall be protected against trivial unauthorized reuse, and inactive sessions shall follow an approved timeout policy.
NFR-SEC-005	Passwords shall never be stored as plaintext; a suitable password-hashing mechanism shall be used.
NFR-SEC-006	Protected clinical information shall not be unnecessarily exposed in logs, errors, URLs, or client-side messages.
NFR-SEC-007	Sensitive actions shall be auditable by actor and timestamp.
NFR-SEC-008	The system shall support appropriate separation of clinical privileges so that reception users cannot perform provider-only clinical actions.

3.4.2 Data Integrity and Clinical Record Safety

ID	Requirement
NFR-DAT-001	Patient identifiers and core record relationships shall remain unique and internally consistent.
NFR-DAT-002	Related changes that must succeed together shall be handled atomically where the persistence technology supports transactions.
NFR-DAT-003	Finalized document content shall be protected against silent modification.
NFR-DAT-004	Amendment history shall remain associated with the original document.
NFR-DAT-005	Required-field and relationship validation shall occur before persistence.
NFR-DAT-006	Patient, appointment, encounter, and document relationships shall prevent accidental cross-patient association.
NFR-DAT-007	Historical appointment and clinical-document states shall remain reconstructable to the extent required by approved policy.

3.4.3 Performance Requirements

ID	Requirement
NFR-PERF-001	Under normal MVP deployment conditions, at least 95% of common read operations such as patient search, chart opening, and appointment-list loading should complete within 2 seconds.
NFR-PERF-002	Under normal MVP deployment conditions, at least 95% of standard patient, appointment, and encounter save operations should complete within 3 seconds, excluding unavailable external services.
NFR-PERF-003	The MVP should support at least 10 concurrent authenticated users without violating the response targets above under a representative workload.

NFR-PERF-004	Performance tests shall document the test environment and workload so that the stated targets are meaningful and repeatable.

3.4.4 Reliability and Failure Handling

ID	Requirement
NFR-REL-001	A temporary network or persistence failure shall result in a clear failure state rather than an unverified success state.
NFR-REL-002	The application shall avoid duplicate creation when a retried operation can be identified as a retry of the same request.
NFR-REL-003	Recoverable errors shall provide a practical next step without exposing implementation details or sensitive information.
NFR-REL-004	Critical data shall be recoverable from backups according to the deployment's approved recovery procedure.
NFR-REL-005	The system shall preserve already-confirmed data when a subsequent operation fails.

3.4.5 Usability and Accessibility

ID	Requirement
NFR-USE-001	Common workflows shall use consistent navigation, terminology, and action placement.
NFR-USE-002	Patient identity and context shall remain visible enough to reduce wrong-patient actions.
NFR-USE-003	Required fields and validation errors shall be understandable to non-technical users.
NFR-USE-004	Appointment states such as scheduled, checked-in, cancelled, and rescheduled shall be visually distinguishable.
NFR-USE-005	Draft and finalized clinical documents shall be visually distinguishable.
NFR-USE-006	Interactive controls should support keyboard navigation and readable labels where the selected UI technology permits.

3.4.6 Maintainability and Extensibility

ID	Requirement
NFR-MNT-001	Business rules shall be separated from presentation concerns sufficiently to allow testing without relying exclusively on UI automation.
NFR-MNT-002	Domain concepts and persistence mappings shall be documented well enough for another developer to understand the model.

NFR-MNT-003	The architecture shall permit future modules such as laboratory, pharmacy, or patient-facing functionality without rewriting the core patient/encounter concepts.
NFR-MNT-004	Requirements, architecture decisions, and implementation changes shall remain traceable through project documentation and version control.
NFR-MNT-005	Significant architectural decisions shall be recorded as ADRs with context, decision, alternatives, consequences, and evidence.

3.4.7 Interoperability and FHIR Readiness

ID	Requirement
NFR-INT-001	The internal clinical model shall maintain clear patient, encounter, appointment, and document concepts that can later be mapped to interoperability standards.
NFR-INT-002	External interoperability shall not require the internal database schema to be a direct copy of an external standard.
NFR-INT-003	API boundaries should be designed so future FHIR-oriented adapters can be added without coupling the UI directly to FHIR payloads.
NFR-INT-004	Production-grade external FHIR integration is outside the MVP unless separately approved.

3.4.8 Auditability

ID	Requirement
NFR-AUD-001	Audit records shall identify the relevant user or system actor and event timestamp.
NFR-AUD-002	Auditable events shall include significant changes to patient records, appointments, clinical documents, patient status, and access/security events as defined by the final audit policy.
NFR-AUD-003	Audit information shall not be editable through ordinary clinical workflows.
NFR-AUD-004	The audit mechanism shall not unnecessarily store sensitive clinical content when event metadata is sufficient.

3.4.9 Backup and Recovery

ID	Requirement
NFR-BAK-001	The deployment shall define a backup mechanism for persistent clinical data before production use.
NFR-BAK-002	Backup and restore procedures shall be documented and tested.

NFR-BAK-003	Backup access shall be protected because backups contain sensitive information.
NFR-BAK-004	Recovery testing shall demonstrate that a representative dataset can be restored without losing required relationships.

3.5 Logical Database Requirements

The logical data model shall represent the following core concepts without prescribing a particular database engine.

Entity	Required logical purpose
Patient	Identity, demographics, identifier, status, and longitudinal patient context.
Allergy	Patient-associated allergy information used for clinical context and alerts.
Medication	Relevant medication information associated with the patient and, where applicable, an encounter.
Appointment	Scheduled patient interaction with date/time, status, service/provider context, and history.
Visit	Operational attendance/check-in context, including walk-in or appointment origin.
Encounter	Clinical interaction containing assessment and care documentation.
Vital Sign	Time-associated clinical measurements recorded for a patient/encounter.
Clinical Note	Narrative documentation associated with an encounter.
Diagnosis	Clinical assessment associated with an encounter and authorized clinician.
Treatment	Treatment/care-plan information associated with an encounter.
Prescription	Prescription information associated with an encounter and authorized clinician.
Clinical Document	Report/note document with lifecycle state and ownership/linkage.
Document Amendment	Traceable correction/addition linked to a finalized original document.
User	Authenticated application identity with one approved role.
Audit Event	Security/clinical activity record containing actor, timestamp, action, and relevant target metadata.

Core relationships shall include: one patient may have many appointments; one patient may have many visits and encounters; an appointment may result in one or more operational events according to workflow policy but shall not be silently duplicated; an encounter belongs to one patient; clinical documents belong to the relevant patient and encounter; amendments reference the finalized document they amend; audit events reference the relevant actor and target where appropriate.

The logical model shall enforce referential integrity and shall distinguish current state from historical state where the business rules require history preservation. Exact field types, indexes, normalization strategy, and physical schema belong to the data-design stage and shall be derived from these requirements.

3.6 Design Constraints

1. The MVP shall use exactly three approved application roles: Clinician/Doctor, Nurse/Clinical Staff, and Receptionist/Front Desk.
2. The MVP shall not include an autonomous AI diagnostic or treatment function.
3. The product shall not treat finalized clinical documents as ordinary mutable comments.
4. The implementation shall preserve patient and encounter relationships as first-class integrity constraints.
5. The design shall support future interoperability without requiring production FHIR integration in the MVP.
6. The system shall remain modular enough for later clinical modules.
7. Security and privacy controls are requirements, not optional enhancements.

3.7 System Attributes and Quality Characteristics

The quality profile of the MVP is summarized below.

Attribute	Priority	Expected quality
Security	Critical	Unauthorized users cannot access protected clinical functions/data.
Privacy	Critical	Sensitive information is disclosed only to authorized users and necessary system components.
Data integrity	Critical	Patient, appointment, encounter, and document relationships remain correct.
Clinical record safety	Critical	Finalized records are preserved and corrections are traceable.
Reliability	High	Failures are visible and do not falsely report successful persistence.
Usability	High	Core workflows are understandable and consistent for all three roles.
Performance	High	Defined response targets are met under the representative MVP workload.
Maintainability	High	Business rules and domain logic remain understandable and testable.
Extensibility	High	Future modules can be added without destabilizing the MVP domain.

Interoperability readiness	Medium	Data/API boundaries allow future standards-based integration.

4 Appendices

The appendices are **supporting requirements material**, not independent requirements. They explain models, evidence, and traceability. If an appendix conflicts with a normative requirement in Section 3, the approved requirement in Section 3 takes precedence until the SRS is formally revised.

4.1 Appendix A — Data Structure Specification

The following logical structures summarize the minimum information expected by the requirements. They are intentionally implementation-neutral.

Structure	Representative attributes
Patient	Patient ID; name; date of birth; sex/gender fields as approved; contact information; address; emergency/contact information where required; status; date of death when applicable; created/updated metadata.
Appointment	Appointment ID; patient ID; date; start/end time; provider/service; status; reason; created by; rescheduled-from reference where applicable; cancellation information.
Visit	Visit ID; patient ID; appointment reference if any; arrival/check-in time; visit type; operational status.
Encounter	Encounter ID; patient ID; visit ID; clinician; start/end time; complaint; history; examination; diagnosis; treatment; prescription reference; note/document references.
Clinical Document	Document ID; patient ID; encounter ID; document type; content/reference; author; created time; finalized time; lifecycle state; version/amendment linkage.
Amendment	Amendment ID; original document ID; author; timestamp; reason; amendment content/reference; approval/status metadata.
Audit Event	Event ID; actor/user; action; target type/ID; timestamp; outcome; relevant metadata.

4.2 Appendix B — Data-Flow and Workflow Models

The conceptual data flow is:

User → Authentication/Authorization → Application Services → Clinical Domain Operations → Persistent Data + Audit

The clinical workflow is:

1. Register or identify the patient.

2. Open the patient chart.
3. Create or retrieve an appointment, or initiate a walk-in.
4. Check in the patient.
5. Create/access the encounter.
6. Record permitted vitals and preliminary information.
7. Clinician performs assessment and records clinical findings.
8. Clinician records diagnosis, treatment, prescription, and notes as applicable.
9. Generate and/or finalize a clinical document.
10. Preserve the resulting encounter/document in the longitudinal patient record.
11. If a finalized document needs correction, create a traceable amendment while preserving the original.

4.3 Appendix C — Detailed Domain/Object Model

The domain model shall treat **Patient** as a central aggregate/context and shall preserve explicit links to **Appointment**, **Visit**, and **Encounter**. The encounter is the principal clinical context for assessment and care documentation. Clinical documents are linked to both patient and relevant encounter so that documents cannot become detached free-floating records.

Important lifecycle concepts include:

- Appointment: scheduled → checked-in/attended, cancelled, or rescheduled as applicable.
- Visit: initiated by appointment or walk-in → checked-in/active → completed according to workflow.
- Encounter: active → completed/closed according to clinical workflow.
- Clinical document: draft → finalized → amended when correction is authorized.
- Patient: active/current status, with a deceased status that affects future scheduling.

4.4 Appendix D — Role and Permission Matrix

Capability	Receptionist / Front Desk	Nurse / Clinical Staff	Clinician / Doctor
Register patient	Yes	Yes, if permitted	Yes, if permitted
Search/view patient	Yes, limited context	Yes	Yes
Manage appointments	Yes	Yes, if permitted	Yes, if permitted
Check-in	Yes	Yes	Yes, if permitted
Walk-in intake	Yes	Yes	Yes, if permitted
Record vitals	No	Yes	Yes

Record diagnosis	No	No unless explicitly authorized by approved policy	Yes
Record treatment	No	No unless explicitly authorized by approved policy	Yes
Record prescription	No	No unless explicitly authorized by approved policy	Yes
Create clinical note	No	Only if explicitly authorized by approved clinical policy	Yes
Generate/finalize medical report	No	Only if explicitly authorized by approved clinical policy	Yes
Amend finalized document	No	Only if explicitly authorized by approved clinical policy	Yes
Change deceased status	No	No unless explicitly approved by clinical policy	Yes
View audit information	Only if explicitly permitted	Only if explicitly permitted	Only if explicitly permitted

This matrix is a requirements baseline, not a claim that every permission has already been clinically approved. The final permission matrix requires mentor/clinical stakeholder confirmation.

4.5 Appendix E — Requirements Traceability Summary

The project's traceability chain is:

Research/Clinical Discovery → Workflow/Use Case → Requirement → Business Rule → Architecture/Data Design → Verification

The principal workflow/use-case references are:

ID	Use case
UC-01	Patient registration and identification
UC-02	Patient search, chart, and clinical summary
UC-03	Appointment scheduling and management

UC-04	Check-in and walk-in handling
UC-05	Clinical encounter and assessment
UC-06	Clinical documentation and reporting
UC-07	Finalization, correction, and amendment of clinical documents
UC-08	Patient status and deceased-patient lifecycle
UC-09	Authentication, authorization, and audit

The complete requirement-to-source mapping is maintained in `docs/03-requirements/traceability-matrix.md`. Verification references are planned evidence; a requirement shall only be considered verified when objective test, review, or acceptance evidence exists.

4.6 Appendix F — Clinical Validation Evidence

The current clinical validation evidence is preliminary. One survey response was obtained from a Doctor/Clinician. Because the sample size is one, the response is treated as qualitative preliminary validation rather than statistical validation.

The response supports the following requirement directions: a useful patient chart should include demographics, conditions, previous diagnoses, vitals, status, allergies, medications, history, and previous visits; appointment permissions may be assigned to nursing/clinical staff; walk-in workflow should remain flexible; cancelled and rescheduled appointments should remain visible in history; encounter data should cover complaint, vitals, history, allergies, medication, examination, diagnosis, treatment, and prescription; clinical notes, diagnosis, treatment, reports, and patient status require clinical authorization; finalized-document mistakes should preserve the original and record a correction/amendment; clinically relevant alerts should be visible; deceased status should include date of death and affect future appointments; and network failures are a practical concern.

These findings have informed the requirements but do not replace broader stakeholder validation. Future review should confirm terminology, exact role permissions, required fields, amendment policy, retention rules, appointment-state behavior, and clinical safety expectations.

4.7 Appendix G — Open Questions and Items for Mentor Review

ID	Open question / decision required
OQ-001	Confirm the final permission matrix for Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
OQ-002	Confirm which patient demographic fields are mandatory for the MVP.
OQ-003	Confirm the exact amendment workflow, including required reason, review, and authorization rules.
OQ-004	Confirm appointment states and whether no-show is included in the MVP lifecycle.
OQ-005	Confirm the policy for future appointments when a patient is marked deceased: cancellation, flagging, or another controlled state.
OQ-006	Confirm minimum audit events and who may view audit information.
OQ-007	Confirm retention and deletion rules for clinical data.

OQ-008	Confirm backup frequency and acceptable recovery objectives for the intended deployment.
OQ-009	Confirm whether allergies and medications can be entered by Nurse/Clinical Staff or require Clinician/Doctor authorization.
OQ-010	Confirm the required report/document types for the MVP.

4.8 Appendix H — Requirements Change Control

Changes to approved requirements shall be recorded with a requirement identifier, reason for change, evidence/source, affected requirements, and review status. Requirements shall not be silently modified in implementation documentation. If a change affects the architecture or data model, the corresponding ADR or design artifact shall also be updated.

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